DRUGS · SPECIAL ENT AGENTS

Drugs for angioedema

Angioedema of the lips, tongue, floor of mouth and larynx is an ENT airway emergency. Its treatment depends on the mechanism. Histamine-mediated (allergic) angioedema, often with urticaria and sometimes with anaphylaxis, responds to adrenaline, antihistamines and steroids. Bradykinin-mediated angioedema, caused by ACE inhibitors or hereditary C1-esterase inhibitor deficiency, does not respond to those drugs and needs airway management and, where available, specific agents such as icatibant or C1-esterase inhibitor concentrate.

1 CHAPTER ONE At a glance

The drugs on one page.

1.1 Summary

DRUGS FOR ANGIOEDEMA

	SUMMARY
Allergic (histaminergic)	Intramuscular adrenaline if there is airway or circulatory compromise (see adrenaline); non-sedating antihistamines; corticosteroids; nebulised adrenaline for laryngeal oedema
ACE inhibitor angioedema	Airway first; stop the ACE inhibitor permanently; antihistamines, steroids and adrenaline are usually ineffective; icatibant (30 mg subcutaneously) and C1-esterase inhibitor or fresh frozen plasma have been used, with mixed trial evidence
Hereditary angioedema (C1-INH deficiency)	Acute: C1-esterase inhibitor concentrate (for example 20 units/kg intravenously), icatibant 30 mg subcutaneously, or ecallantide; prophylaxis: lanadelumab, berotralstat, C1-INH; short-term prophylaxis before surgery and dental work (C1-INH, or danazol or tranexamic acid in some protocols)

	SUMMARY
Key features of bradykinin angioedema	No urticaria or itch; slower onset over hours; lasts 2 to 5 days; abdominal attacks in hereditary disease; family history; normal tryptase; low C4 in hereditary types

2 CHAPTER TWO Using them in ENT

Practice.

2.1 ACE inhibitor angioedema

It affects about 0.1% to 0.7% of patients on ACE inhibitors, more often people of African descent, and can occur after years of uneventful treatment. It typically involves the tongue, lips and floor of mouth. Assess the airway with flexible nasendoscopy; the patient's voice, drooling, stridor and the extent of swelling (tongue base and larynx) decide whether to secure the airway, which may need awake fibreoptic intubation or a surgical airway. Admit patients with floor of mouth, tongue or laryngeal involvement. Switching to an angiotensin receptor blocker carries a small risk of recurrence.

2.2 Hereditary angioedema

Diagnosed by low C4 and low C1-esterase inhibitor level or function; triggered by trauma, surgery, dental work, oestrogens and ACE inhibitors. Before ENT surgery, arrange short-term prophylaxis with immunology.

3 CHAPTER THREE Safety

What goes wrong.

3.1 Key points

Giving only antihistamines and steroids to a patient with bradykinin angioedema and an evolving airway problem delays the decisions that matter. Icatibant causes injection-site reactions; C1-esterase inhibitor carries a small thrombosis risk.

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- How to tell **histaminergic from bradykinin** angioedema.
- Management of **ACE inhibitor angioedema**, starting with the airway.
- Hereditary angioedema: diagnosis and perioperative prophylaxis.

4.2 Common pitfalls

COMMON PITFALLS

- Discharging a patient with tongue swelling on an ACE inhibitor after antihistamines.
- Restarting the ACE inhibitor.

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